

ADVANCED

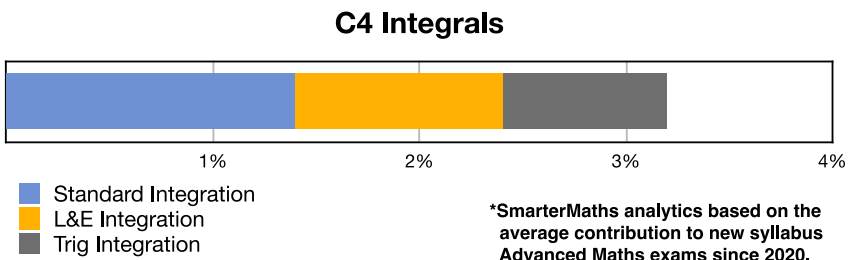
Stage 6

Calculus (Adv), C4 Integration (Adv), Integration (Y12)

Standard Integration

Teacher: SHS Maths

Exam Equivalent Time: 63 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- C4 Integration* has contributed a substantial 14.0% to new syllabus Advanced exams since it was introduced in 2020.
- This topic has been split into four sub-topics for analysis purposes: *1-Integrals (3.2%)*, *2-Areas Under Curves (5.8%)*, *3-Trapezoidal Rule (2.2%)* and *4-Other Integration Examples (2.8%)*.
- This analysis looks at *Integrals*.

HSC ANALYSIS - What to expect and common pitfalls

- Integrals (3.2%)* is made up of standard integration, log and exponential integration and trig integration.
- Dedicated questions testing standard integration have appeared in every new syllabus exam between the 2021-24, including 2 separate questions in 2022. Before this, it has appeared sporadically with a volatile mark allocation (including a notable 6 mark allocation in 2009).
- Standard integrals are a great opportunity to score highly, although attention should be given to *2011 Adv 2e* and *2011 Adv 4d*, which produced sub-50% mean marks.
- A "differentiate then integrate" question was asked in 2022 within this sub-topic and should be part of any revision.

Questions

1. Calculus, 2ADV C4 2024 HSC 5 MC

What is $\int (6x + 1)^3 dx$?

- A. $\frac{1}{24}(6x + 1)^4 + C$
- B. $\frac{1}{4}(6x + 1)^4 + C$
- C. $\frac{2}{3}(6x + 1)^4 + C$
- D. $\frac{3}{2}(6x + 1)^4 + C$

2. Calculus, 2ADV C4 SM-Bank 4 MC

Let $f'(x) = \frac{2}{\sqrt{2x - 3}}$.

If $f(6) = 4$, then

- A. $f(x) = 2\sqrt{2x - 3}$
- B. $f(x) = \sqrt{2x - 3} - 2$
- C. $f(x) = 2\sqrt{2x - 3} - 2$
- D. $f(x) = \sqrt{2x - 3} + 2$

3. Calculus, 2ADV C4 2022 HSC 6 MC

What is $\int \frac{1}{(2x + 1)^2} dx$?

- A. $\frac{-2}{2x + 1} + C$
- B. $\frac{-1}{2(2x + 1)} + C$
- C. $2\ln(2x + 1) + C$
- D. $\frac{1}{2}\ln(2x + 1) + C$

4. Calculus, 2ADV C4 SM-Bank 5

Let $f'(x) = x^3 + x$.

Find $f(x)$ given that $f(1) = 2$. (2 marks)

5. Calculus, 2ADV C4 2016 HSC 11d

Evaluate $\int_0^1 (2x + 1)^3 \, dx$. (2 marks)

6. Calculus, 2ADV C4 2017 HSC 11b

Find $\int (2x + 1)^4 \, dx$. (1 mark)

7. Calculus, 2ADV C4 2009 HSC 2b

i. Find $\int 5 \, dx$. (1 mark)

ii. Find $\int \frac{3}{(x - 6)^2} \, dx$. (2 marks)

iii. Evaluate $\int_1^4 x^2 + \sqrt{x} \, dx$. (3 marks)

8. Calculus, 2ADV C4 2022 HSC 18

a. Differentiate $y = (x^2 + 1)^4$. (2 marks)

b. Hence, or otherwise, find $\int x(x^2 + 1)^3 \, dx$. (1 mark)

9. Calculus, 2ADV C4 2007 HSC 2bii

Evaluate $\int_1^4 \frac{8}{x^2} \, dx$. (3 marks)

10. Calculus, 2ADV C4 2010 HSC 2e

Given that $\int_0^6 (x + k) \, dx = 30$, and k is a constant, find the value of k . (2 marks)

11. Calculus, 2ADV C4 2019 MET1 2

Find $f(x)$ given that $f(1) = -\frac{7}{4}$ and $f'(x) = 2x^2 - \frac{1}{4}x^{-\frac{2}{3}}$. (2 marks)

12. Calculus, 2ADV C4 2021 HSC 15

Evaluate $\int_{-2}^0 \sqrt{2x + 4} \, dx$. (2 marks)

13. Calculus, 2ADV C4 2023 HSC 17

Find $\int x\sqrt{x^2 + 1} \, dx$ (2 marks)

14. Calculus, 2ADV C4 2010 HSC 2di

Find $\int \sqrt{5x + 1} \, dx$. (2 marks)

15. Calculus, 2ADV C4 2019 HSC 11e

Evaluate $\int_0^1 \frac{1}{(3x + 2)^2} \, dx$. (2 marks)

16. Calculus, 2ADV C4 2014 HSC 11d

Find $\int \frac{1}{(x + 3)^2} \, dx$. (2 marks)

17. Calculus, 2ADV C4 2011 HSC 4d

i. Differentiate $y = \sqrt{9 - x^2}$ with respect to x . (2 marks)

ii. Hence, or otherwise, find $\int \frac{6x}{\sqrt{9 - x^2}} \, dx$. (2 marks)

18. Calculus, 2ADV C4 2011 HSC 2e

Find $\int \frac{1}{3x^2} \, dx$. (2 marks)

Worked Solutions

1. Calculus, 2ADV C4 2024 HSC 5 MC

$$\begin{aligned}\int (6x+1)^3 dx &= \frac{1}{4} \cdot \frac{1}{6} (6x+1)^4 + C \\ &= \frac{1}{24} (6x+1)^4 + C \\ &\Rightarrow A\end{aligned}$$

2. Calculus, 2ADV C4 SM-Bank 4 MC

$$\begin{aligned}f'(x) &= \frac{2}{\sqrt{2x-3}} \\ f(x) &= 2 \int (2x-3)^{-\frac{1}{2}} \\ &= 2 \cdot \frac{1}{2} \cdot 2(2x-3)^{\frac{1}{2}} + c \\ &= 2\sqrt{2x-3} + c\end{aligned}$$

When $x = 6$, $f(x) = 4$:

$$\begin{aligned}4 &= 2\sqrt{12-3} + c \Rightarrow c = -2 \\ \therefore f(x) &= 2\sqrt{2x-3} - 2 \\ &\Rightarrow C\end{aligned}$$

3. Calculus, 2ADV C4 2022 HSC 6 MC

$$\begin{aligned}\int (2x+1)^{-2} &= \frac{(2x+1)^{-1}}{(-1)(2)} + C \\ &= \frac{-1}{2(2x+1)} + C \\ &\Rightarrow B\end{aligned}$$

Mean mark 53%.
COMMENT: A poor State
result warrants attention.

Worked Solutions

4. Calculus, 2ADV C4 SM-Bank 5

$$\begin{aligned}f'(x) &= x^3 + x \\ f(x) &= \int x^3 + x \, dx \\ &= \frac{1}{4}x^4 + \frac{1}{2}x^2 + c\end{aligned}$$

Given $f(1) = 2$:

$$\begin{aligned}2 &= \frac{1}{4} + \frac{1}{2} + c \\ c &= \frac{5}{4}\end{aligned}$$

$$\therefore f(x) = \frac{1}{4}x^4 + \frac{1}{2}x^2 + \frac{5}{4}$$

5. Calculus, 2ADV C4 2016 HSC 11d

$$\begin{aligned}\int_0^1 (2x+1)^3 \, dx &= \frac{1}{4} \cdot \frac{1}{2} \left[(2x+1)^4 \right]_0^1 \\ &= \frac{1}{8} [3^4 - 1^4] \\ &= \frac{1}{8} \times 80 \\ &= 10\end{aligned}$$

6. Calculus, 2ADV C4 2017 HSC 11b

$$\begin{aligned}\int (2x+1)^4 \, dx &= \frac{1}{5} \times \frac{1}{2} \times (2x+1)^5 + C \\ &= \frac{(2x+1)^5}{10} + C\end{aligned}$$

7. Calculus, 2ADV C4 2009 HSC 2b

i. $\int 5 \, dx = 5x + C$

ii.
$$\begin{aligned} \int \frac{3}{(x-6)^2} \, dx \\ &= 3 \int (x-6)^{-2} \, dx \\ &= 3 \times \frac{1}{-1} \times (x-6)^{-1} + c \\ &= \frac{-3}{(x-6)} + c \end{aligned}$$

iii.
$$\begin{aligned} \int_1^4 x^2 + \sqrt{x} \, dx \\ &= \int_1^4 \left(x^2 + x^{\frac{1}{2}} \right) \, dx \\ &= \left[\frac{1}{3} x^3 + \frac{\frac{3}{2}}{\frac{3}{2}} x^{\frac{3}{2}} \right]_1^4 \\ &= \left[\frac{x^3}{3} + \frac{2}{3} x^{\frac{3}{2}} \right]_1^4 \\ &= \left[\left(\frac{4^3}{3} + \frac{2}{3} \times 4^{\frac{3}{2}} \right) - \left(\frac{1}{3} + \frac{2}{3} \right) \right] \\ &= \left[\left(\frac{64}{3} + \frac{16}{3} \right) - \frac{3}{3} \right] \\ &= \left[\frac{80}{3} - \frac{3}{3} \right] \\ &= \frac{77}{3} \end{aligned}$$

8. Calculus, 2ADV C4 2022 HSC 18

a. $y = (x^2 + 1)^4$

Using chain rule:

$$\begin{aligned} \frac{dy}{dx} &= 4 \times 2x (x^2 + 1)^3 \\ &= 8x (x^2 + 1)^3 \end{aligned}$$

b.
$$\begin{aligned} \int x (x^2 + 1)^3 \, dx &= \frac{1}{8} \int 8x (x^2 + 1)^3 \, dx \\ &= \frac{1}{8} (x^2 + 1)^4 + C \end{aligned}$$

9. Calculus, 2ADV C4 2007 HSC 2bii

$$\begin{aligned} \int_1^4 \frac{8}{x^2} \, dx \\ &= 8 \int_1^4 x^{-2} \, dx \\ &= 8 \left[-\frac{1}{x} \right]_1^4 \\ &= 8 \left[\left(-\frac{1}{4} \right) - \left(-\frac{1}{1} \right) \right] \\ &= 8 \left(\frac{3}{4} \right) \\ &= 6 \end{aligned}$$

10. Calculus, 2ADV C4 2010 HSC 2e

$$\begin{aligned} \int_0^6 (x + k) \, dx &= 30 \\ \int_0^6 (x + k) \, dx &= \left[\frac{1}{2} x^2 + kx \right]_0^6 \\ &= \left[\left(\frac{1}{2} \times 6^2 + 6 \times k \right) - 0 \right] \\ &= 18 + 6k \end{aligned}$$

$$\Rightarrow 18 + 6k = 30$$

$$6k = 12$$

$$\therefore k = 2$$

11. Calculus, 2ADV C4 2019 MET1 2

$$\begin{aligned} f(x) &= \int 2x^2 - \frac{1}{4}x^{-\frac{2}{3}} dx \\ &= \frac{2}{3}x^3 - \frac{1}{4} \cdot \frac{1}{\frac{1}{3}}x^{\frac{1}{3}} + c \\ &= \frac{2}{3}x^3 - \frac{3}{4}x^{\frac{1}{3}} + c \end{aligned}$$

Given $f(1) = -\frac{7}{4}$:

$$\begin{aligned} -\frac{7}{4} &= \frac{2}{3} - \frac{3}{4} + c \\ c &= -\frac{5}{3} \end{aligned}$$

$$\therefore f(x) = \frac{2}{3}x^3 - \frac{3}{4}x^{\frac{1}{3}} - \frac{5}{3}$$

12. Calculus, 2ADV C4 2021 HSC 15

$$\begin{aligned} \int_{-2}^0 (2x+4)^{\frac{1}{2}} dx &= \left[\frac{2}{3} \cdot \frac{1}{2} (2x+4)^{\frac{3}{2}} \right]_{-2}^0 \\ &= \frac{1}{3} \left(4^{\frac{3}{2}} - 0 \right) \\ &= \frac{8}{3} \end{aligned}$$

13. Calculus, 2ADV C4 2023 HSC 17

$$\begin{aligned} \int x\sqrt{x^2+1} dx &= \frac{1}{2} \int 2x(x^2+1)^{\frac{1}{2}} dx \\ &= \frac{1}{2} \times \frac{2}{3} (x^2+1)^{\frac{3}{2}} + c \\ &= \frac{1}{3} (x^2+1)^{\frac{3}{2}} + c \end{aligned}$$

Mean mark 52%.

14. Calculus, 2ADV C4 2010 HSC 2di

$$\begin{aligned} \int \sqrt{5x+1} dx &= \frac{1}{\frac{3}{2}} \times \frac{1}{5} \times (5x+1)^{\frac{3}{2}} + C \\ &= \frac{2}{15} (5x+1)^{\frac{3}{2}} + C \end{aligned}$$

15. Calculus, 2ADV C4 2019 HSC 11e

$$\begin{aligned} \int_0^1 \frac{1}{(3x+2)^2} dx &= \int_0^1 (3x+2)^{-2} dx \\ &= \left[-\frac{1}{3} (3x+2)^{-1} \right]_0^1 \\ &= \left[-\frac{1}{3} \cdot \frac{1}{5} - \left(-\frac{1}{3} \cdot \frac{1}{2} \right) \right] \\ &= -\frac{1}{15} + \frac{1}{6} \\ &= \frac{1}{10} \end{aligned}$$

16. Calculus, 2ADV C4 2014 HSC 11d

$$\begin{aligned} \int \frac{1}{(x+3)^2} dx &= \int (x+3)^{-2} dx \\ &= \frac{1}{-1} \cdot (x+3)^{-1} + C \\ &= \frac{-1}{(x+3)} + C \end{aligned}$$

17. Calculus, 2ADV C4 2011 HSC 4d

i. $y = \sqrt{9-x^2}$
 $= (9-x^2)^{\frac{1}{2}}$

IMPORTANT: Some students might find calculations easier by rewriting the equation as $y = (9-x^2)^{\frac{1}{2}}$.

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} \times (9-x^2)^{-\frac{1}{2}} \times \frac{d}{dx} (9-x^2) \\ &= \frac{1}{2} \times (9-x^2)^{-\frac{1}{2}} \times -2x \\ &= -\frac{x}{\sqrt{9-x^2}} \end{aligned}$$

ii. $\int \frac{6x}{\sqrt{9-x^2}} dx = -6 \int \frac{-x}{\sqrt{9-x^2}} dx$
 $= -6 \left(\sqrt{9-x^2} \right) + C$
 $= -6\sqrt{9-x^2} + C$

18. Calculus, 2ADV C4 2011 HSC 2e

$$\begin{aligned}\int \frac{1}{3x^2} \, dx &= \frac{1}{3} \int x^{-2} \, dx \\ &= \frac{1}{3} \times \frac{1}{-1} \times x^{-1} + C \\ &= -\frac{1}{3x} + C\end{aligned}$$

◆ Mean mark 46%.

MARKER'S COMMENT: Students who took the $\frac{1}{3}$ out the front before integrating made less errors.